# SVD Weaving Draft Explorer — Tutorial

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## Introduction

This page lets you take any black-and-white bitmap image, factor it into three matrices using Singular Value Decomposition (SVD), and display the result as a standard weaving draft — threading, tie-up, treadling, and drawdown. You can then colour the warp and weft threads, preview the woven cloth, and export the draft as a WIF file for use in weaving software such as Fiberworks or PixeLoom.

The page is designed for students learning about matrix factorisation in a textile context. The central idea is that a binary image (a grid of 0s and 1s) is a matrix, and SVD gives a principled way to factor any matrix into three pieces. In weaving, those three pieces correspond exactly to the structure of a draft: which shafts each warp thread passes through (threading), which treadles lift which shafts (tie-up), and which treadle is pressed on each pick (treadling).

**What you need:** A black-and-white or greyscale image saved as a BMP file. Most image editors (GIMP, Photoshop, Paint) can save to BMP. The image should be relatively small — 20×20 to 100×100 pixels works well. Very large images will be slower to compute.

## Step 1: Upload a BMP image

At the top of the page, click the upload area (or drag and drop a file onto it) to load a BMP image. The page accepts BMP files at any bit depth — 1-bit (pure black and white), 8-bit greyscale, 24-bit colour, or 32-bit colour with transparency. Colour images are automatically converted to greyscale and then thresholded: pixels below 50% brightness become 1 (dark/warp-up), and pixels above become 0 (light/weft-up).

Once the file loads, two pieces of information appear:

- **X** — the width of the image in pixels, which will become the number of warp threads in the draft
- **Y** — the height of the image in pixels, which will become the number of picks (weft rows)

A status line confirms the filename and pixel dimensions.

## Step 2: Scale the image

Below the dimension display, a **Scale** field (default 100%) lets you shrink the image before computing the draft. Enter any integer from 1 to 100 and click Apply.

Scaling uses nearest-neighbour resampling — each output pixel takes the value of the nearest input pixel. The original full-resolution image is always kept in memory, so you can move the scale up and down freely without losing quality. Setting the scale to 50% on a 60×80 image gives a 30×40 working matrix; setting it back to 100% restores all 60×80 pixels exactly.

**Why scale?** Smaller matrices produce faster SVD computation and smaller drafts that are easier to read. A 10×10 image produces a draft that fits comfortably on screen at any Z. Very large images (200+ pixels

in either dimension) may take several seconds to compute, especially with the Hungarian algorithm enabled.

## Step 3: Edit the matrix

Below the scale controls, the image is displayed as a grid of small squares — black for 1 (dark pixel, warp up) and white for 0 (light pixel, warp down). This is the binary matrix that will be factored by SVD.

You can edit this matrix directly:

- **Click** any cell to toggle it between 0 and 1
- **Click and drag** to paint a stroke — the value you paint is the opposite of whatever you clicked first, so dragging across white cells turns them black, and dragging across black cells turns them white, without accidentally reversing cells you have already painted

Edits apply to the current scaled matrix only. If you change the scale after editing, the edits are lost — the scale slider always resamples from the original uploaded image. Edit after scaling, not before.

This editor lets you draw simple patterns directly if you don't have a BMP file ready, or correct small details in an uploaded image before computing the draft.

## Step 4: Set Z and compute the draft

**Z** is the number of shafts (and treadles) in the resulting draft. Enter a value in the Shafts / treadles field and click **Create draft through matrix factorization**.

Z controls the complexity of the approximation:

- **Small Z (e.g. 2–4):** The draft is simple — few shafts, a coarse approximation. The drawdown will look quite different from the original image, capturing only the broadest structure.
- **Large Z (e.g. 20–30):** More shafts and treadles, finer approximation. The drawdown will resemble the original image more closely.
- **Z = min(image width, image height):** Full rank. Given the right assignment method (see Step 4a), the drawdown can perfectly reconstruct the original image.

Z cannot exceed the smaller of the image's two dimensions — the SVD of a Y×X matrix can produce at most min(Y, X) meaningful components. If you request a larger Z, the extra shafts are filled with zeros and shown as inactive rows in the threading and treadling.

After computing, a status line confirms the actual dimensions of all four draft matrices.

## Understanding the draft

The draft is displayed in four panels arranged in the conventional weaving layout.

**Threading (Z shafts × X warps) — shown in blue, top left** Each column represents one warp thread. The filled cell in that column shows which shaft the thread is on. Every warp thread is assigned to exactly one shaft, which matches the physical constraint of a real loom.

**Tie-up (Z × Z) — shown in green, top right** Rows are shafts (top = shaft 1 at the top of the threading), columns are treadles (left = treadle 1). A filled cell means pressing that treadle lifts that shaft. The tie-up is derived from the original image given the threading and treadling assignments: each cell is 1 if the majority of pixels where that shaft/treadle combination applies are dark.

**Treadling (Y picks × Z treadles) — shown in red, below the tie-up** Each row represents one pick (weft row). The filled cell shows which treadle is pressed on that pick. Like the threading, each pick is assigned to exactly one treadle.

**Drawdown (Y picks × X warps) — shown in black, bottom left** The drawdown is the reconstructed image computed from the three matrices: for each cell (pick y, warp x), if there is any shaft i such that warp x is threaded on shaft i, shaft i is tied to a treadle j that pick y presses, and tieup[i][j] = 1, then the cell is filled (warp is up). The drawdown approximates the original image, with error coming from the quantisation of continuous SVD values into binary draft matrices.

## The assignment methods: argmax vs. Hungarian algorithm

Once the SVD gives continuous singular vectors, each warp thread must be assigned to exactly one shaft and each pick to exactly one treadle. The page offers two methods, selectable via the checkbox before computing.

**Argmax (default, faster)** Each warp is assigned to the shaft whose singular vector has the largest absolute value at that position, independently of all other warps. This is fast but greedy — multiple warps can end up on the same shaft, leaving other shafts empty. When that happens, the tie-up block for that shaft covers several pixels and must represent them with a single 0 or 1, losing detail.

**Hungarian algorithm (slower, more accurate)** Solves the globally optimal 1-to-1 assignment problem: finds the assignment of warps to shafts (and picks to treadles) that maximises the total alignment with the singular vectors, subject to the constraint that no two warps share a shaft. This is available when $Z \geq \min(\text{image width}, \text{image height})$. When the assignment is a true permutation, each tie-up cell covers exactly one pixel from the original image, and the drawdown can reconstruct the image perfectly. The algorithm runs in $O(n^3)$ time and will be noticeably slower on large images.

The status bar after computation confirms which method was used and whether it succeeded or fell back to argmax due to insufficient rank.

## Step 5: Colour schemes

Below the draft, two colour panels let you assign colours to warp threads and weft picks independently.

**Preset schemes** Click any preset button to apply a scheme instantly. Presets use repeating patterns — alternating pairs, stripes of 4, gradients — applied across all threads at once. The active preset is highlighted.

**Custom scheme builder** Click the ＋ **Create custom scheme** link to expand the builder. Choose up to three colours (A, B, C) using the colour pickers. Then choose a pattern — All A, Alternate A/B, Stripe 4, Stripe 3, Gradient — using the pattern buttons. A preview strip shows how the colours will be distributed

across the threads. Type a name and click ＋ **Add** to save it as a new preset button. The draft must be computed before you can save a custom scheme.

**Individual thread overrides** Below the scheme buttons, a row of small colour swatches shows the current colour of every thread or pick. Click any swatch to open a 40-colour palette popup. Click a colour to apply it, or use the Custom picker at the bottom of the popup for any hex colour. Individual overrides deselect the active preset.

**Coloured drawdown (Section 7)** Below the colour panels, a canvas renders the woven cloth in colour. Each cell shows the warp colour when the warp is up (filled in the drawdown) and the weft colour when the weft is up. The canvas updates live whenever any colour changes. A thin colour strip along the bottom edge shows the warp colour sequence.

## Step 6: Download the WIF file

At the bottom of the page, click ⬇ **Download WIF file** to save the draft in Weaving Information File (WIF 1.1) format. This file can be opened directly in Fiberworks PCW, PixeLoom, or any other WIF-compatible weaving software.

The WIF file includes:

- **[WEAVING]** — shaft and treadle count (both equal to Z)
- **[THREADING]** — one line per warp thread, listing its shaft assignment
- **[TIEUP]** — one line per shaft, listing the treadles it is tied to
- **[TREADLING]** — one line per pick, listing its treadle assignment
- **[COLOR TABLE]** — all unique colours currently assigned to warp and weft threads, as RGB values
- **[WARP COLORS]** and **[WEFT COLORS]** — the colour index for each individual thread and pick

Orientation note: the WIF format numbers warp threads right-to-left relative to the on-screen display, shafts bottom-to-top, and picks top-to-bottom. These flips are applied automatically so the draft loads correctly in Fiberworks. If the orientation looks wrong in your software, check whether it uses rising or falling shed convention — the file is written as a rising shed draft.

The Download WIF button is disabled until a draft has been computed.

## Step 7: Notes and downloading your notes

Section 8 of the page has a three-tab notes area.

**Observations tab** Six guided questions prompt you to describe the original image, compare the drawdown to the original at different values of Z, observe how the threading and tie-up change, compare the two assignment methods, and describe your colour choices. Type your answers in the text boxes.

**Experiments tab** A list of suggested activities (set Z = 1, set Z to the full rank, draw a simple shape in the matrix editor, scale the image before factoring) with a matching "what to look for" column explaining the underlying mathematics.

**My Notes tab** A free-form text area for any other observations.

**Downloading your notes** Click ⬇ **Download notes** at the bottom of the page to save a plain text file. The file includes your image dimensions and Z, the full warp and weft colour assignments for every thread, an ASCII-art representation of the tie-up, all six observation answers, and your freeform notes. This gives a complete record of one run of the activity that can be submitted or reviewed later.

Both download buttons are grouped together at the very bottom of the page and are disabled until a draft has been computed.

## The mathematics: SVD and weaving (optional reading)

A binary image is a matrix M of shape Y×X, where each entry is 0 or 1. Singular Value Decomposition factors any matrix as:

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M = U \Sigma V^T
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where U is Y×r, Σ is a diagonal r×r matrix of non-negative singular values (in descending order), and V^T is r×X. Here r = min(Y, X) is the rank of the matrix.

Each singular value $\sigma_i$ and its corresponding columns $u_i$ (from U) and $v_i$ (from V) describe one "layer" of the matrix — a rank-1 component. The full matrix is the sum of r such layers: $M = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \ldots + \sigma_r u_r v_r^T$. Larger singular values capture more of the image's energy; keeping only the top Z terms gives the best rank-Z approximation in the least-squares sense.

**How this maps to a weaving draft:**

- **V^T (right singular vectors, r×X)** — each row is a pattern across the X warp threads. This becomes the threading: each warp is assigned to the shaft (row of V^T) where its value is largest.
- **U (left singular vectors, Y×r)** — each column is a pattern across the Y picks. This becomes the treadling: each pick is assigned to the treadle (column of U) where its value is largest.
- **Σ (singular values, diagonal r×r)** — the tie-up is derived from which shaft/treadle combinations correspond to dark pixels in the original image.

**Why Z matters:** Keeping only Z singular values is a lossy compression. Small Z ≈ coarse approximation; Z = min(Y, X) ≈ exact reconstruction (given the right assignment). The difference between the drawdown and the original image is the *quantisation error* introduced by forcing continuous SVD values into the binary constraints of a loom: each thread must be on exactly one shaft, and each pick must press exactly one treadle.

**The Hungarian algorithm** solves the assignment problem optimally — finding the permutation of shafts and treadles that minimises total reconstruction error — rather than assigning each thread greedily. At full rank with optimal assignment, the draft is an exact factorisation of the original binary image.